1.

Suppose u is a non-negative harmonic function on $B(0,R)$.
(1) Use the Poisson formula to prove

$$R^{n-2}\frac{R-|x|}{(R+|x|)^{n-1}}u(0)\leqslant u(x)\leqslant R^{n-2}\frac{R+|x|}{(R-|x|)^{n-1}}u(0);$$

(2) Prove the following Harnack's inequality: for each $r\in(0,R)$,

$$\sup_{B(0,r)}u\leqslant\left(\frac{R+r}{R-r}\right)^n\inf_{B(0,r)}u.$$

Proof: (1) By Poisson formula, for $r\in(|x|,R)$,

$$u(x)=\int_{\partial B(0,r)}K(x,y)u(y)\,\mathrm{d}s(y),\quad\text{where }K(x,y)=\frac{r^2-|x|^2}{n\alpha(n)r|x-y|^n}.$$

Hence $K(x,y)\geqslant\frac{r^2-|x|^2}{n\alpha(n)(r+|x|)^nr}=\frac{r-|x|}{n\alpha(n)(r+|x|)^{n-1}r}$, and

$$u(x)\geqslant\frac{r-|x|}{n\alpha(n)(r+|x|)^{n-1}r}\int_{\partial B(0,r)}u(y)\,\mathrm{d}s(y)=\frac{r-|x|}{(r+|x|)^{n-1}}r^{n-2}u(0).$$

Let $r\to R$ we have

$$u(x)\geqslant R^{n-2}\frac{R-|x|}{(R+|x|)^{n-1}}u(0)$$

and analogously $u(x)\leqslant R^{n-2}\frac{R+|x|}{(R-|x|)^{n-1}}u(0)$.
(2) By maximum principle, the sup and inf are taken at $\partial B(0,r)$, so by (1) $\sup u\leqslant R^{n-2}\frac{R+r}{(R-r)^{n-1}}u(0)$ $\leqslant\left(\frac{R+r}{R-r}\right)^n\inf u$.

2.

Prove using the previous exercise: if u is harmonic on $\mathbb{R}^n$ and bounded from above, then u is constant.
Proof: If $u\leqslant M$ then $v=M-u$ is a non-negative harmonic function on $\mathbb{R}^n$. By 1(2) for any $0<r<R$ we have $\sup_{B(0,r)}v\leqslant\left(\frac{R+r}{R-r}\right)^n\inf_{B(0,r)}v$, so let $R\to\infty$ we have $\sup_{B(0,r)}v=\inf_{B(0,r)}v$. Hence v is constant, and so is u.

3.

Suppose $\Omega=B^+(R)=\{x\in B(0,R):x_n>0\}$. Determine the Green function of the equation

$$\begin{cases} -\Delta u = f, & x \in \Omega \\ u = g, & x \in \partial\Omega \end{cases}$$

Solution: First suppose $R = 1$. Recall for $x \in \mathbb{R}^n$, $\bar{x} = (x_1, \cdots, x_{n-1}, -x_n)$ and $x^* = x/|x|^2$. Consider $G(x, y) = \Gamma(y - x) - \Gamma(y - \bar{x}) - \Gamma(|x|(y - x^*)) + \Gamma(|x|(y - \bar{x}^*))$ for $y \in \overline{\Omega}$ and $x \in \Omega$. Then clearly $y \neq \bar{x}, x^*, \bar{x}^*$, so $-\Delta_y G(x, y) = -\Delta_y \Gamma(y - x) = \delta_0(y - x)$ if $y \in \Omega$. Note that $y \in \partial\Omega$ consists of two parts: either $|y| = 1$ or $y_n = 0$. If $|y| = 1$, then $\Gamma(y - x) = \Gamma(|x|(y - x^*))$ and $\Gamma(y - \bar{x}) = \Gamma(|x|(y - \bar{x}^*))$; otherwise if $y_n = 0$, then $\Gamma(y - x) = \Gamma(y - \bar{x})$, and $\Gamma(|x|(y - x^*)) = \Gamma(|x|(y - \bar{x}^*))$. Hence for any $y \in \partial\Omega$ we have $G(x, y) = 0$, so $G(x, y)$ is indeed a Green function on Ω.
For general R, analogously the Green function is

$$G(x, y) = \Gamma(y - x) - \Gamma(y - \bar{x}) - \Gamma\left(\frac{|x|}{R}(y - x^*)\right) + \Gamma\left(\frac{|x|}{R}(y - \bar{x}^*)\right).$$

4.

Suppose $\Omega \subset \mathbb{R}^n$ is bounded, $u \in C(\overline{\Omega}) \cap C^2(\Omega)$ is the solution to

$$\begin{cases} -\Delta u = 1, & x \in \Omega \\ u = 0, & x \in \partial\Omega \end{cases}$$

Prove that for any $x_0 \in \Omega$,

$$\frac{1}{2n} \min_{x \in \partial\Omega} |x - x_0|^2 \leqslant u(x_0) \leqslant \frac{1}{2n} \max_{x \in \partial\Omega} |x - x_0|^2.$$

Proof: Let $v(x) = \frac{1}{2n}|x - x_0|^2$, then $\Delta v = 1$ for $x \in \Omega$, so $u + v$ is a harmonic function on Ω. By maximum principle,

$$u(x_0) = (u + v)(x_0) \leqslant \max_{\overline{\Omega}} (u + v) = \max_{\partial\Omega} (u + v) = \frac{1}{2n} \max_{x \in \partial\Omega} |x - x_0|^2.$$

and likewise considering minimum we have the other side of the desired inequality.

5.

Suppose $\Omega \subset \mathbb{R}^n$ is bounded and open ($n \geqslant 3$) with smooth boundary, consider

$$\begin{cases} -\Delta u + A \cdot Du + cu = f, & x \in \Omega \\ u = 0, & x \in \partial\Omega \end{cases} \tag{$\star$}$$

where $A : \Omega \to \mathbb{R}^n$ and $c : \Omega \to \mathbb{R}$ is continuous and bounded. If $c(x) - \frac{1}{4}|A(x)|^2 > 0$, prove the solution is unique, using energy estimates.
Proof: Suppose there are two solutions u_1, u_2, then since the equation is linear, $u = u_1 - u_2$ satisfy the equation

$$\begin{cases} -\Delta u + A \cdot Du + cu = 0, & x \in \Omega \\ u = 0, & x \in \partial\Omega \end{cases}$$

Analogous to the energy estimate, integration by parts gives

$$\int_\Omega (-\Delta u) \cdot u \, \mathrm{d}x = \int_\Omega |Du|^2 \, \mathrm{d}x - \int_\Omega \operatorname{div}(u \nabla u) \, \mathrm{d}x = \int_\Omega |Du|^2 \, \mathrm{d}x$$

and

$$\int_\Omega (-\Delta u) \cdot u \, \mathrm{d}x = \int_\Omega (-A \cdot Du - cu) u \, \mathrm{d}x.$$

Therefore

$$\int_\Omega |Du|^2 + (A \cdot Du) u + cu^2 \, \mathrm{d}x = 0.$$

If $c(x) - \frac{1}{4}|A(x)|^2 > 0$, then on any compact set $K \subset \Omega$, $c - \frac{1}{4}A^2 > \varepsilon$ for some $\varepsilon > 0$, so $|Du|^2 + (A \cdot Du)u + cu^2 \geqslant \varepsilon u^2$, and $0 \geqslant \varepsilon \int_K u^2 \, \mathrm{d}x$ which implies $u|_K = 0$. Therefore $u = 0$, so $u_1 = u_2$ and the solution to $(\star)$ is unique.

6.

Suppose $\Omega \subset \mathbb{R}^n$ is bounded and open ($n \geqslant 3$), $x_0 \in \partial\Omega$. If $u \in C^2(\Omega) \cap C(\overline{\Omega} \setminus \{x_0\})$ is a bounded solution to

$$\begin{cases} -\Delta u = f, & x \in \Omega \\ u = g, & x \in \partial\Omega \setminus \{x_0\} \end{cases}$$

Prove that such a solution is unique.
Proof: It suffices to show that the only solution to $-\Delta u = 0, x \in \Omega$ and $u = 0, x \in \partial\Omega \setminus \{x_0\}$ such that $u \in C^2(\Omega) \cap C(\overline{\Omega} \setminus \{x_0\})$ is zero.
Recall that $\Delta|x - x_0|^{2-n} = 0$, so for small ε, consider $u_\varepsilon(x) = \varepsilon|x - x_0|^{2-n} + u(x)$, then $-\Delta u_\varepsilon = 0 \forall x \in \Omega$ and $u_\varepsilon = \varepsilon|x - x_0|^{2-n} \geqslant \varepsilon d^{2-n}$. Suppose u is bounded by M, then for small enough r such that $\varepsilon r^{2-n} \geqslant M$, we have $u_e(x) \geqslant 0 \forall x \in B(x_0, r)$. Let $\Omega^\varepsilon = \Omega \setminus \overline{B(x_0, r)}$, then by minimum principle

$$\min_{x \in \overline{\Omega^\varepsilon}} u_\varepsilon(x) = \min_{x \in \partial\overline{\Omega^\varepsilon}} u_\varepsilon(x) \geqslant 0$$

Since $u_\varepsilon \to u$ and $u_\varepsilon \geqslant 0 \forall x \in \overline{\Omega^\varepsilon}$, we have $u \geqslant 0 \forall x \in \overline{\Omega} \setminus \{x_0\}$. Analogously $u \leqslant 0$, so $u \equiv 0$.